Sangwoo **Seo**

Postdoctoral Researcher @ LG AI Research.

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Education

KAIST (Korea Advanced Institute of Science and Technology) — *Daejeon, South Korea*

PH.D in Industrial & Systems Engineering — *Aug. 2022 - Aug. 2026*

- Advisor: Prof. Chanyoung Park
- Research Interest: Graph Neural Networks

KAIST (Korea Advanced Institute of Science and Technology) — *Daejeon, South Korea*

M.S in Electrical Engineering — *Aug. 2020 - Aug. 2022*

- Advisor: Prof. Changick Kim
- Research Interest: Graph Neural Networks

DGIST (Daegu Gyeongbuk Institute of Science and Technology) — *Daegu, South Korea*

B.S. in School of Undergraduate Studies — *Mar. 2014 - Aug. 2020*

- Major in Electrical Engineering

Position

Materials Intelligence Lab, LG AI Research — *Seoul, South Korea*

Postdoctoral Researcher — *Aug. 2026 - Present*

- Research area: Development of a Materials Foundation Model for Integrating Organic and Inorganic Materials

Graph and Geometric Learning Lab, Yale University — *New Haven, CT, USA*

Visiting Researcher — *Nov. 2025 - Apr. 2026*

- Advisor: Rex Ying
- Research area: Circuit graph foundation model based on multimodal

Research Interests

Graph Neural Networks

- Explainability in GNN
- Multimodal learning in GNN

AI for Science, AI for engineering

- Physics AI
- Mesh Graph Simulation
- Circuit GNN

Publication

- **ELASTIC: Trajectory-Based Synchronization of Event and Tracking Data in Soccer** — *2026*
  Hyunsung Kim, Hoyoung Choi, Kunhee Lee, Sangwoo Seo, Tom Boomstra, Jinsung Yoon, Chanyoung Park
  ACM International Conference on Information and Knowledge Management (CIKM)

- **Bridging Circuit Graphs and Language: A Cross-Modal Transformer for Netlist-to-Text Alignment and Generation** — *2026*
  Sangwoo Seo, Hyunsung Kim, Hyunchul Kim, Byeonggon Kang, Rex Ying, Chanyoung Park
  ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) Workshop on Graph Machine Learning for the Large Model Era

- **Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations** — *2026*
  Sangwoo Seo, Hyunsung Kim, Jiwan Kim, Chanyoung Park
  ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) Workshop on Reliable Scientific Foundation Models

- **PathCRF: Ball-Free Soccer Event Detection via Possession Path Inference from Player Trajectories** *2026*
  Hyunsung Kim, Kunhee Lee, Sangwoo Seo, Sangki Ko, Jinsung Yoon, Chanyoung Park
  ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)

- **CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs** *2026*
  Jiwan Kim, Kibum Kim, Sangwoo Seo, Chanyoung Park
  International Conference on Learning Representations (ICLR)

- **Better Prevent than Tackle: Valuing Defense in Soccer Based on Graph Neural Networks** *2026*
  Hyunsung Kim, Sangwoo Seo, Hoyoung Choi, Tom Boomstra, Jinsung Yoon, Chanyoung Park
  MIT Sloan Sports Analytics Conference (SSAC)

- **Target Circuit Matching in Large-Scale Netlists using GNN-Based Region Prediction** *2025*
  Sangwoo Seo, Jimin Seo, Yoonho Lee, Donghyeon Kim, Hyejin Shin, Banghyun Sung, Chanyoung Park
  International Conference on Computer-Aided Design (ICCAD)

- **Disentangling Hyperedges through the lens of Category Theory** *2025*
  Yoonho Lee, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, Chanyoung Park
  Conference on Neural Information Processing Systems (NeurIPS)

- **ELASTIC: Event-Tracking Data Synchronization in Soccer Without Annotated Event Locations** *2025*
  Hyunsung Kim, Hoyoung Choi, Sangwoo Seo, Tom Boomstra, Jinsung Yoon, Chanyoung Park
  European Conference on Machine Learning and Principles and Practice Knowledge Discovery in Databases (ECML-PKDD) Workshop on MLSA

- **Global Context-aware Representation Learning for Spatially Resolved Transcriptomics** *2025*
  Yunhak Oh, Junseok Lee, Yeongmin Kim, Sangwoo Seo, Namkyeong Lee, Chanyoung Park
  International Conference on Machine Learning (ICML)

- **Capturing functional context of genetic pathways through hyperedge disentanglement** *2025*
  Yoonho Lee, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, Chanyoung Park
  International Conference on Learning Representations (ICLR) Workshop on Machine Learning for Genomics Explorations

- **Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck** *2024*
  Sangwoo Seo, Sungwon Kim, Jihyeong Jung, Yoonho Lee, and Chanyoung Park
  ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)

- **Interpretable Graph Model with Prototype-Based Graph Information Bottleneck** *2024*
  Sangwoo Seo, Sungwon Kim, and Chanyoung Park
  ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) Workshop on Human-Interpretable AI

- **Unsupervised Episode Generation for Graph Meta-learning** *2024*
  Jihyeong Jung, Sangwoo Seo, Sungwon Kim, and Chanyoung Park
  International Conference on Machine Learning (ICML)

- **Interpretable Prototype-based Graph Information Bottleneck** *2023*
  Sangwoo Seo, Sungwon Kim, and Chanyoung Park
  Conference on Neural Information Processing Systems (NeurIPS) **[Gold Prize at the 30th Samsung Humantech Paper Award]**

- **Explanation-based Graph Neural Networks for Graph Classification** *2022*
  Sangwoo Seo, Seungjun Jung, and Changick Kim
  International Conference on Pattern Recognition (ICPR)

Fellowships and Awards

2024 **Best Paper Award at KDD 2024 Workshop on Human-Interpretable AI**, ACM KDD *Barcelona, Spain*
2024 **KDD 2024 Student Travel Award** , ACM KDD *Barcelona, Spain*
2023 **Gold Prize at the 30th Samsung Humantech Paper Award** [KRW 20M] , Samsung Electronics *Suwon, S. Korea*

Reviewer Service

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**): 2024, 2026
- Conference on Neural Information Processing Systems (**NeurIPS**): 2024. 2025, 2026
- International Conference on Learning Representations (**ICLR**): 2025
- Association for the Advancement of Artificial Intelligence (**AAAI**): 2026
- International Conference on Artificial Intelligence and Statistics (**AISTATS**): 2025, 2026, 2027
- International Conference on Machine Learning (**ICML**): 2025, 2026
- Conference on Information and Knowledge Management (**CIKM**): 2025, 2026
- Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**): 2025

Experience

Materials Intelligence Lab, LG AI Research — *Seoul, South Korea*
PROJECT PARTICIPANT — *Aug. 2026 - present*
- Partner: POSTECH
- Developments in AI Foundation Models for Property Prediction of Industrial Materials

Graph and Geometric Learning Lab, Yale University — *New Haven, CT, USA*
PROJECT PARTICIPANT — *Nov. 2025 - Apr. 2026*
- Advisor: Rex Ying
- RAG system for financial reports

Data Science & Artificial Intelligence Lab, KAIST — *Daejeon, South Korea*
PROJECT PARTICIPANT — *Mar. 2024 - Nov. 2024*
- Partner: SK Hynix
- Development of a circuit recognition technique using graph neural networks

Computational Intelligence Lab, KAIST — *Daejeon, South Korea*
PROJECT PARTICIPANT — *Aug. 2020 - Aug. 2022*
- Partner: Satrec Initiative Analytics
- Method design for eliminating internal bias in graph neural networks

Information Communications & Security Lab, DGIST — *Daegu, South Korea*
RESEARCH INTERNSHIP — *Jan. 2018 - Feb. 2018*
- Advisor: Prof. Jaemin Lee
- A theoretical modeling on mobile edge computing based on queuing theory and stochastic geometry

Integrated NanoSystems Lab, DGIST — *Daegu, South Korea*
RESEARCH INTERNSHIP — *Jun. 2017 - Aug. 2017*
- Advisor: Prof. Junghyup Lee
- Circuit design and production

Invited Talk

- **Top Conference Session: Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck** — *Yeosu, South Korea*
  KOREA SOFTWARE CONGRESS (KSC) — *Dec. 2024*
- **Flash Talk: Interpretable Prototype-based Graph Information Bottleneck** — *Jeju, South Korea*
  Y-KAST INTERNATIONAL CONFERENCE — *Mar. 2024*
- **Top Conference Session: Interpretable Prototype-based Graph Information Bottleneck** — *Busan, South Korea*
  KOREA SOFTWARE CONGRESS (KSC) — *Dec. 2023*

Teaching Experience

- **Teaching Assistant in Samsung Electronics**
  Data Science Specialist Course (Jun. 2024)

- **Teaching Assistant in Samsung Electronics**
  AI Specialist Course (Sep. 2023)
- **Teaching Assistant in KAIST**
  DS503: Machine Learning for Data Science (Spring 2023)
- **Instructor in Cantho University**
  Instruction in Programming Languages including Python and C (Summer 2015)

References

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Rex Ying, Ph.D.
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